Bridge Life Cycle Cost Analysis Report

Project information

Project name:	D_35_2L_OF_P
Project code:	D_35_2L_OF_P
Project description:	Darbhanga	35m	2 Lane	One side Footpath	PSC Bridge
Prepared using:	3psLCCA (osdag.iitb.ac.in/3pslcca)

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1 Introduction to Life Cycle Cost Assessment

Life Cycle Cost is the total cost incurred by the bridge throughout its life. This report presents Life Cycle Cost Assessment (LCCA) based on the Three Pillars of Sustainability-Life Cycle Cost (3PS-LCC) framework, which considers the economic, social, and environmental pillars over the selected analysis period. The various life-cycle stages and sustainability pillars considered in the 3PS-LCC framework are illustrated in Figure 1.

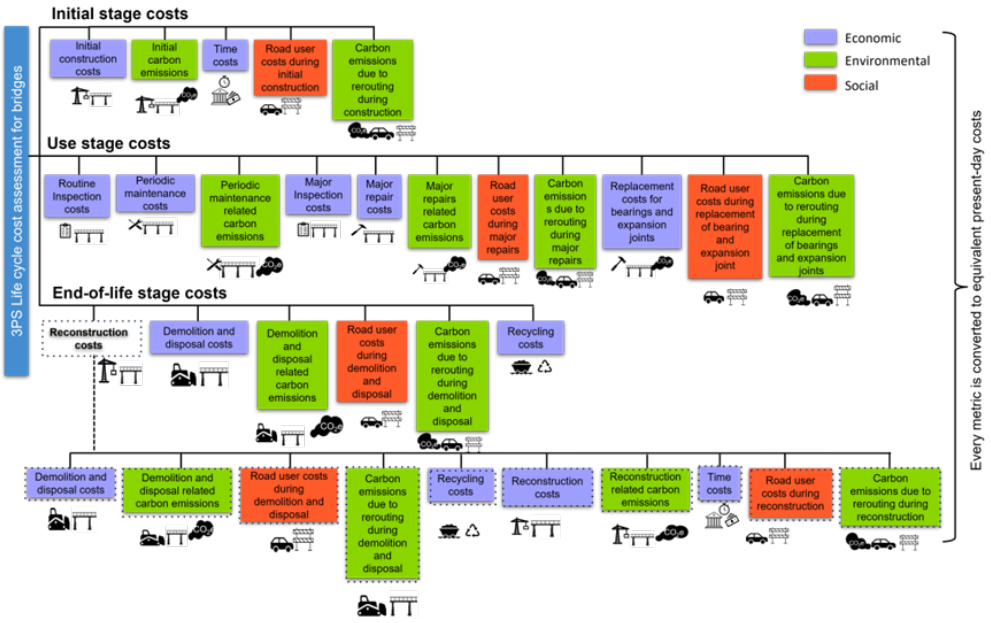


Figure 1. 3PS-LCC framework

2 Input data

2.1 Bridge geometry and description

Table 1. Bridge Data Summary

Bridge Identification	
Name of the Bridge	D_35_2L_OF_P
Owner	—
Location	
Country	India
Bridge Alignment & Location	Darbhanga
Technical Specifications	
Type of Bridge	Girder
Span	35.0 (m)
Carriageway Width	9.0 (m)
Number of Lanes	2
Vehicle Path Direction	One Way
Footpath	Footpath at one side
Life Cycle	
Design Life	100 (years)
Analysis Period	100 (years)
Year of Construction	2025
Construction Schedule	
Duration of Construction	9.2 (months)
Working Days per Month	26 (days)
Days per Month	30 (days)

2.2 Financial inputs

Table 2. Financial Data Summary

Financial Data	
Discount Rate	6.7 (%)
Source: Discount Rate	—
Inflation Rate	5.15 (%)
Source: Inflation Rate	—
Interest Rate	7.75 (%)
Source: Interest Rate	—
Investment Ratio	0.5
Source: Investment Ratio	—

2.3 Construction data

Table 3. Structure Work Data: Foundation

Material	**Quantity**	**Unit**	**Rate/Unit**	**Rate Source**	**Total (INR)**
Excavation					
Excavation for foundation (pile cap depth + depth pf PCC)‡	104.98	m³	107.95	—	11,332.16
Backfilling E#	104.98	m³	0.00	—	0.00
Pile					
Boring and concreting in piles M35†	40.00	m	19,204.70	—	768,188.00
Reinforcement in pile†	20.01	t	74,458.40	—	1,489,912.58
Pile Cap					
Levelling course for Pile Cap (Plain Cement Concrete M15)†	5.62	m³	4,257.10	—	23,946.19
Reinforcement in pile cap†	3.70	t	74,458.40	—	275,496.08
Concreting in pile cap M40#	84.38	m³	5,590.60	—	471,706.88

All rates and totals in INR. — Database value (default); # Manually entered; § Modified from database; † Imported from Excel; ‡ Imported from Excel and modified

Table 4. Structure Work Data: Sub-Structure

Material	Quantity	Unit	Rate/Unit	Rate Source	Total (INR)
Pier					
Reinforcement in pier‡	2.33	t	74,692.40	—	173,734.52
Concreting in pier M40#	11.69	m³	6,187.60	—	72,333.04
Pier Cap					
Reinforcement in pier cap†	2.34	t	74,692.40	—	174,780.22
Concreting in pier cap M40#	23.05	m³	6,187.60	—	142,605.62
Pedestal					
Reinforcement in pedestal‡	0.03	t	74,692.40	—	2,420.03
Concrete in pedestal M40#	0.22	m³	6,187.60	—	1,336.52
Bearings					
Pot-PTFE bearing#	2,412.93	t	213.80	—	515,884.43

All rates and totals in INR. — Database value (default); # Manually entered; § Modified from database; † Imported from Excel; ‡ Imported from Excel and modified

Table 5. Structure Work Data: Super Structure

Material	Quantity	Unit	Rate/Unit	Rate Source	Total (INR)
Girder					
Reinforcement in girder‡	13.88	t	74,926.20	—	1,040,050.58
Prestressing tendons including ancillaries‡	15.17	t	150,364.00	—	2,280,570.79
Concreting in girder M45#	126.19	m³	13,260.00	—	1,673,279.40
Deck Slab					
Concreting in deck slab M45#	84.53	m³	6,912.90	—	584,312.87
Reinforcement in deck slab#	10.14	m³	74,926.20	—	759,976.45
Diaphragm					
Reinforcement in diaphragms†	1.20	t	74,926.20	—	89,557.79
Concreting in diaphragms M45#	11.95	m³	13,022.55	—	155,656.07
Cross Bracings					

All rates and totals in INR. — Database value (default); # Manually entered; § Modified from database; † Imported from Excel; ‡ Imported from Excel and modified

Table 6. Structure Work Data: Miscellaneous

Material	Quantity	Unit	Rate/Unit	Rate Source	Total (INR)
Railing & Crash Barrier & Median					
Steel railing‡	35.00	m	3,758.70	—	131,554.50
Crash barrier (M40)†	70.00	m	4,018.40	—	281,288.00
Median/divider†	35.00	m	3,437.80	—	120,323.00
Drainage					
Asphalt, Utilities and Other Materials					
Strip seal expansion joint†	10.50	m	13,860.90	—	145,539.45
12 mm thick mastic asphalt wearing course†	367.50	m²	486.60	—	178,825.50
Yellow and white paint on road for directions†	3.50	m²	747.80	—	2,617.30
Anti corrosive paint for reinforcement‡	53.62	m	11,287.00	—	605,208.94
Waterproofing					
Waterproofing for pile cap‡	101.25	m²	432.00	—	43,740.00
Waterproofing for deck slab†	367.50	m²	246.00	—	90,405.00

All rates and totals in INR. — Database value (default); # Manually entered; § Modified from database; † Imported from Excel; ‡ Imported from Excel and modified

2.4 Maintenance data

Table 7. Maintenance Data Summary

Routine Maintenance	
Routine Inspection Cost	0.1 (% of initial construction cost)
Routine Inspection Frequency	1 (year)
Periodic Maintenance	
Periodic Maintenance Cost	0.55 (% of initial construction cost)
Periodic Maintenance Carbon Cost	0.55 (% of initial construction cost)
Periodic Maintenance Frequency	5 (year)
Major Inspection	
Major Inspection Cost	0.5 (% of initial construction cost)
Major Inspection Frequency	5 (year)
Major Repair	
Major Repair Cost	10.0 (% of initial construction cost)
Major Repair Carbon Cost	0.55 (% of initial construction cost)
Major Repair Frequency	20 (year)
Major Repair Duration	3 (months)
Bearings & Expansion Joints	
Bearing & Expansion Joint Replacement Cost	12.5 (% of initial construction cost)
Bearing & Expansion Joint Replacement Frequency	25 (year)
Bearing & Expansion Joint Replacement Duration	2 (days)

2.5 Traffic data

2.5.1 Traffic and Road Data

Table 8 presents the traffic and road parameters recorded for this project, including road classification, speed limits, and associated data used in the life cycle cost assessment.

Table 8. Traffic and Road Data

Rerouting Road Configuration	
Rerouting Road Configuration	Two Lane
Carriageway Width	7.0 (m)
Road Hourly Capacity	1900 (PCU/hr)
Accident Severity Distribution	
Minor Injury	1.64 (%)
Major Injury	20.34 (%)
Fatal Accident	78.02 (%)
Road Parameters	
Road Roughness	2000.0 (mm/km)
Road Rise	0.0 (m/km)
Road Fall	0.0 (m/km)
Rerouting Distance	0.175 (km)
Rerouting Time	0.525 (min)
Crash Rate along Rerouting Route	79.01 (acc / M km)
Work Zone Multiplier	1.0
Traffic Flow	
Number of Peak Hours	2

2.6 Environmental input data

2.6.1 Social Cost of Carbon

The social cost of carbon (SCC) quantifies the monetary value of environmental damage caused by each unit of CO_2e emitted. The following section presents the SCC values and parameters adopted in this assessment.

The Social Cost of Carbon (SCC) is entered manually as **8.04 INR/kgCO_2e**. Source: Source not mentioned.

2.6.2 Material Emission Factors

Table 9 lists the materials included in the carbon emissions calculation along with their respective emission factors, quantities, and total CO_2e contributions. Materials excluded from the assessment are detailed in Table 10.

Table 9. Materials Included in Carbon Emissions Calculation

Material	Quantity	Unit	Conversion Factor	Emission Factor	Emission Factor Unit	Total ($kgCO_2e$)
Foundation – Pile						
Boring and concreting in piles M35	40.00	m	4,239.00	0.08	$kgCO_2e/kg$	14,243.04
Reinforcement in pile	20.01	t	1,000.00	2.60	$kgCO_2e/kg$	52,026.00
Foundation – Pile Cap						
Levelling course for Pile Cap (Plain Cement Concrete M15)	5.62	m^3	2,400.00	0.11	$kgCO_2e/kg$	1,485.00
Reinforcement in pile cap	3.70	t	1,000.00	2.60	$kgCO_2e/kg$	9,620.00
Concreting in pile cap M40	84.38	m^3	2,400.00	0.08	$kgCO_2e/kg$	17,010.00
Sub-Structure – Pier						
Reinforcement in pier	2.33	t	1,000.00	2.60	$kgCO_2e/kg$	6,047.60
Concreting in pier M40	11.69	m^3	2,400.00	0.08	$kgCO_2e/kg$	2,356.70
Sub-Structure – Pier Cap						
Reinforcement in pier cap	2.34	t	1,000.00	2.60	$kgCO_2e/kg$	6,084.00
Concreting in pier cap M40	23.05	m^3	2,400.00	0.08	$kgCO_2e/kg$	4,646.28
Sub-Structure – Pedestal						
Reinforcement in pedestal	0.03	t	1,000.00	2.60	$kgCO_2e/kg$	84.24
Concrete in pedestal M40	0.22	m^3	2,400.00	0.08	$kgCO_2e/kg$	43.55
Super-Structure – Girder						
Reinforcement in girder	13.88	t	1,000.00	2.60	$kgCO_2e/kg$	36,090.60
Prestressing tendons including ancillaries	15.17	t	1,000.00	2.16	$kgCO_2e/kg$	32,760.72
Concreting in girder M45	126.19	m^3	2,400.00	0.08	$kgCO_2e/kg$	25,439.90
Super-Structure – Deck Slab						
Concreting in deck slab M45	84.53	m^3	2,400.00	0.08	$kgCO_2e/kg$	17,040.24
Reinforcement in deck slab	10.14	m^3	1,000.00	2.60	$kgCO_2e/kg$	26,371.80
Super-Structure – Diaphragm						
Reinforcement in diaphragms	1.20	t	1,000.00	2.60	$kgCO_2e/kg$	3,107.73

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Table 9. Materials Included in Carbon Emissions Calculation

Material	Quantity	Unit	Conversion Factor	Emission Factor	Emission Factor Unit	Total ($kgCO_2e$)
Concreting in diaphragms M45	11.95	m³	2,400.00	0.08	$kgCO_2e/kg$	2,409.69
Misc – Railing & Crash Barrier & Median						
Steel railing	35.00	m	10.20	2.50	$kgCO_2e/kg$	892.50
Crash barrier (M40)	70.00	m	715.20	0.11	$kgCO_2e/kg$	5,507.04
Misc – Asphalt, Utilities and Other Materials						
12 mm thick mastic asphalt wearing course	367.50	m²	162.00	0.01	$kgCO_2e/kg$	714.42
Yellow and white paint on road for directions	3.50	m²	2.15	0.25	$kgCO_2e/kg$	1.88
Misc – Waterproofing						
Waterproofing for pile cap	101.25	m²	120.00	0.08	$kgCO_2e/kg$	1,020.60
Waterproofing for deck slab	367.50	m²	2.00	0.19	$kgCO_2e/kg$	139.65

Table 10. Materials Excluded from Carbon Emissions Calculation

Material	Exclusion Reason
Foundation – Excavation	
Excavation for foundation (pile cap depth + depth pf PCC)	Incomplete Data
Backfilling E	Incomplete Data
Sub-Structure – Bearings	
Pot-PTFE bearing	Incomplete Data
Misc – Railing & Crash Barrier & Median	
Median/divider	Manually Excluded
Misc – Asphalt, Utilities and Other Materials	
Strip seal expansion joint	Incomplete Data
Anti corrosive paint for reinforcement	Incomplete Data

2.7 Recycling data

Table 11. Materials Included in Recyclability Calculation

Material	Recyclability %	Recyclable Quantity	Unit	Scrap Rate (INR)	Recovered Value (INR)
Foundation – Pile					
Reinforcement in pile	75.00	15.01	t	32,500.00	487,743.75
Foundation – Pile Cap					
Reinforcement in pile cap	75.00	2.78	t	32,500.00	90,187.50
Sub-Structure – Pier					
Reinforcement in pier	75.00	1.74	t	32,500.00	56,696.25
Sub-Structure – Pier Cap					
Reinforcement in pier cap	75.00	1.75	t	32,500.00	57,037.50
Sub-Structure – Pedestal					
Reinforcement in pedestal	75.00	0.02	t	32,500.00	789.75
Super-Structure – Girder					
Reinforcement in girder	75.00	10.41	t	32,500.00	338,349.38
Super-Structure – Deck Slab					
Reinforcement in deck slab	75.00	7.61	m³	32.50	247.24
Super-Structure – Diaphragm					
Reinforcement in diaphragms	75.00	0.90	t	32,500.00	29,134.95

Table 12. Materials Excluded from Recyclability Calculation

Material	Reason
Foundation – Excavation	
Excavation for foundation (pile cap depth + depth pf PCC)	Incomplete Data
Backfilling E	Incomplete Data
Foundation – Pile	
Boring and concreting in piles M35	Incomplete Data
Foundation – Pile Cap	
Levelling course for Pile Cap (Plain Cement Concrete M15)	Incomplete Data

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Table 12. Materials Excluded from Recyclability Calculation

Material	Reason
Concreting in pile cap M40	Incomplete Data
Sub-Structure – Pier	
Concreting in pier M40	Incomplete Data
Sub-Structure – Pier Cap	
Concreting in pier cap M40	Incomplete Data
Sub-Structure – Pedestal	
Concrete in pedestal M40	Incomplete Data
Sub-Structure – Bearings	
Pot-PTFE bearing	Incomplete Data
Super-Structure – Girder	
Prestressing tendons including ancillaries	Manually Excluded
Concreting in girder M45	Incomplete Data
Super-Structure – Deck Slab	
Concreting in deck slab M45	Incomplete Data
Super-Structure – Diaphragm	
Concreting in diaphragms M45	Incomplete Data
Misc – Railing & Crash Barrier & Median	
Steel railing	Manually Excluded
Crash barrier (M40)	Incomplete Data
Median/divider	Incomplete Data
Misc – Asphalt, Utilities and Other Materials	
Strip seal expansion joint	Incomplete Data
12 mm thick mastic asphalt wearing course	Incomplete Data
Yellow and white paint on road for directions	Incomplete Data
Anti corrosive paint for reinforcement	Incomplete Data
Misc – Waterproofing	
Waterproofing for pile cap	Incomplete Data
Waterproofing for deck slab	Incomplete Data

3 LCCA results

3.1 Life cycle cost results

Table 13 presents a comprehensive summary of the life cycle cost analysis results, expressed as present values. The costs are organised by life cycle stage — Initial Stage, Use Stage, Reconstruction, and End-of-Life — and further broken down by sustainability pillar: Economic, Environmental, and Social.

Table 13. Life Cycle Cost Analysis Results

Description	Present Value (INR)
Initial Stage	
Economic	
Initial Construction Cost	12,306,581.91
Time Costs	365,608.04
Environmental	
Initial Carbon Emissions	2,242,217.11
Carbon emissions due to Rerouting (Construction)	1,339,099.99
Social	
Road User Costs (Construction)	31,197,250.69
Stage Total – Initial Stage	**47,450,757.73**
Use Stage	
Economic	
Routine Inspection Costs	631,610.70
Periodic Maintenance Costs	662,106.41
Major Inspection Costs	601,914.92
Major Repair Costs	2,468,700.33
Replacement Costs of Bearings and Expansion joints	1,227,804.84
Environmental	
Periodic Maintenance related Carbon Emissions	120,633.52
Major Repair related Carbon Emissions	24,738.38
Carbon Emissions due to Rerouting during Major Repairs	875,946.06
Carbon Emissions due to Rerouting during Replacement of Bearings and Expansion joints	14,477.81
Social	
Road User Costs during Major Repairs	20,407,071.16
Road User Costs during Replacement of Bearings and Expansion joints	337,292.01
Stage Total – Use Stage	**27,372,296.15**
End-of-Life Stage	
Economic	

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Table 13. Life Cycle Cost Analysis Results

Description	Present Value (INR)
Demolition and Disposal Costs	284,282.04
Recycling Costs	-244,903.04
Environmental	
Demolition and Disposal related Carbon Emissions	51,795.22
Carbon Emissions due to Rerouting during Demolition and Disposal	33,623.05
Social	
Road User Costs due to Demolition and Disposal	783,322.27
Stage Total – End-of-Life Stage	**908,119.54**
Total life cycle cost	**75,731,173.42**

The life cycle cost results are further illustrated through the following figures. Figure 2, Figure 3, Figure 4, Figure 5 present the distribution of costs across the three sustainability pillars (Economic, Environmental, and Social) and the four life cycle stages, providing a visual overview of the relative contributions to the total life cycle cost.

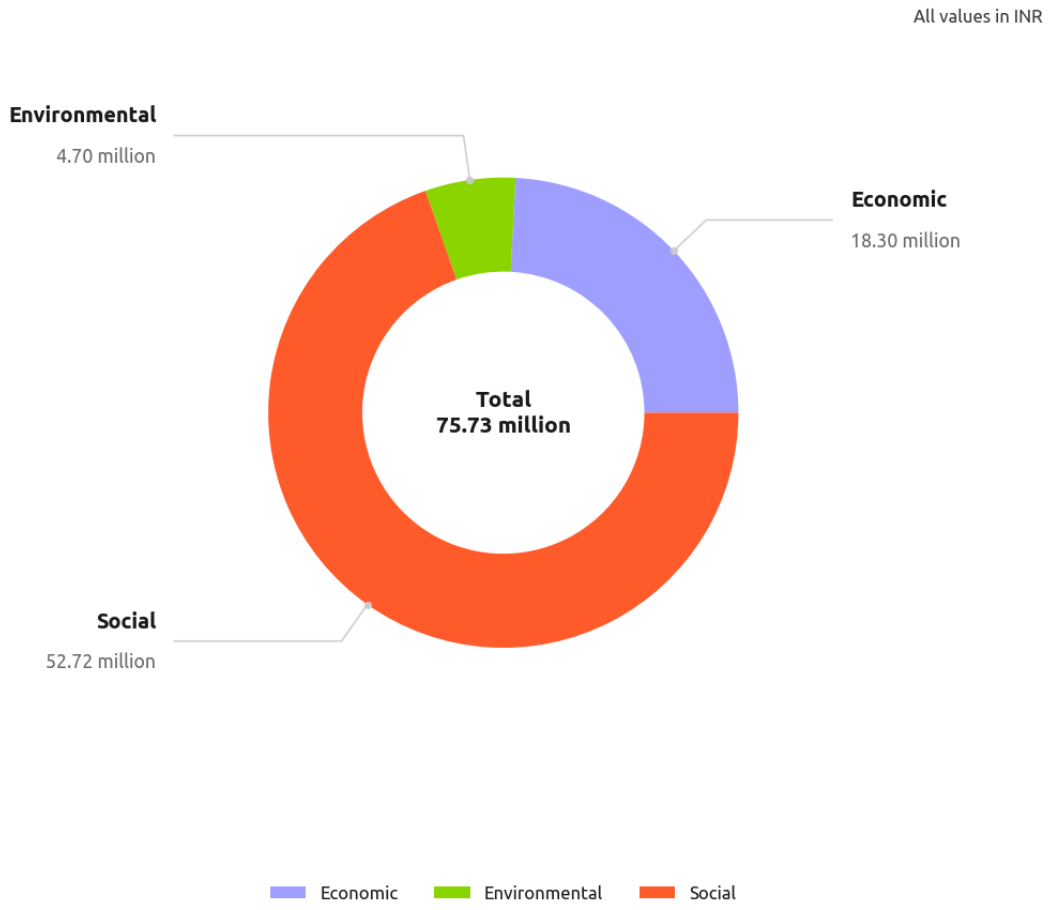


Figure 2. LCC components results

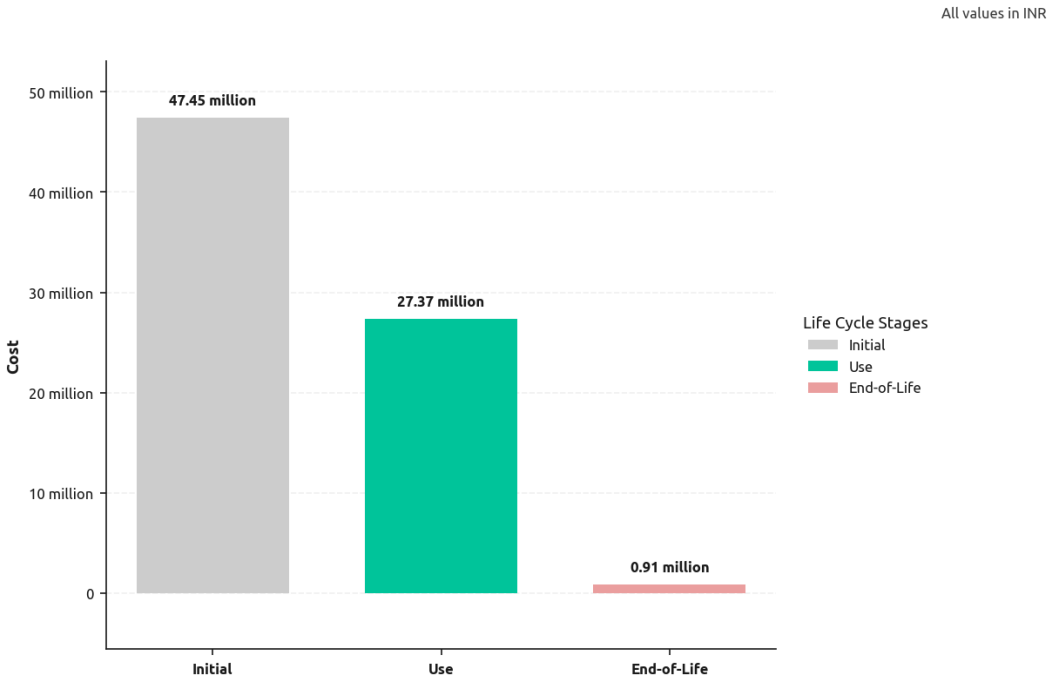


Figure 3. Distribution of 3PS and 3 stages of LCC

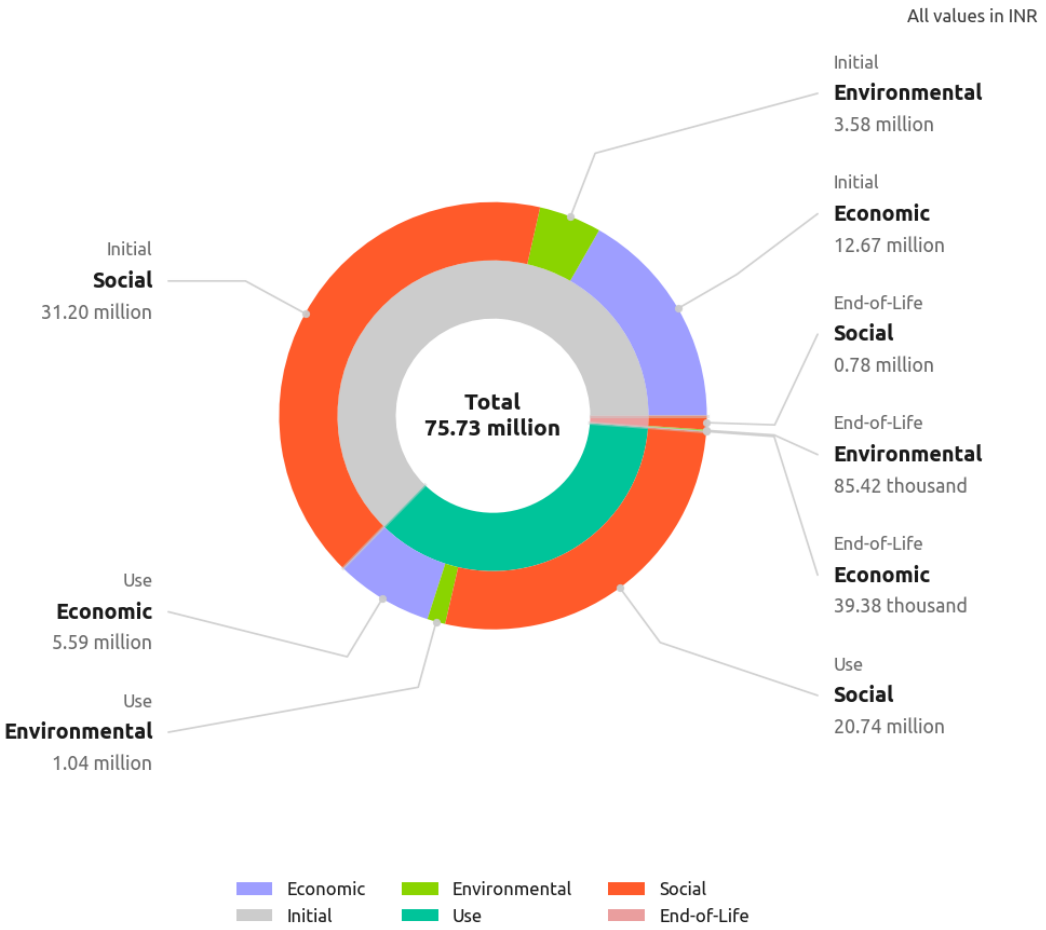


Figure 4. Distribution of 3PS and sustainability pillars of LCC

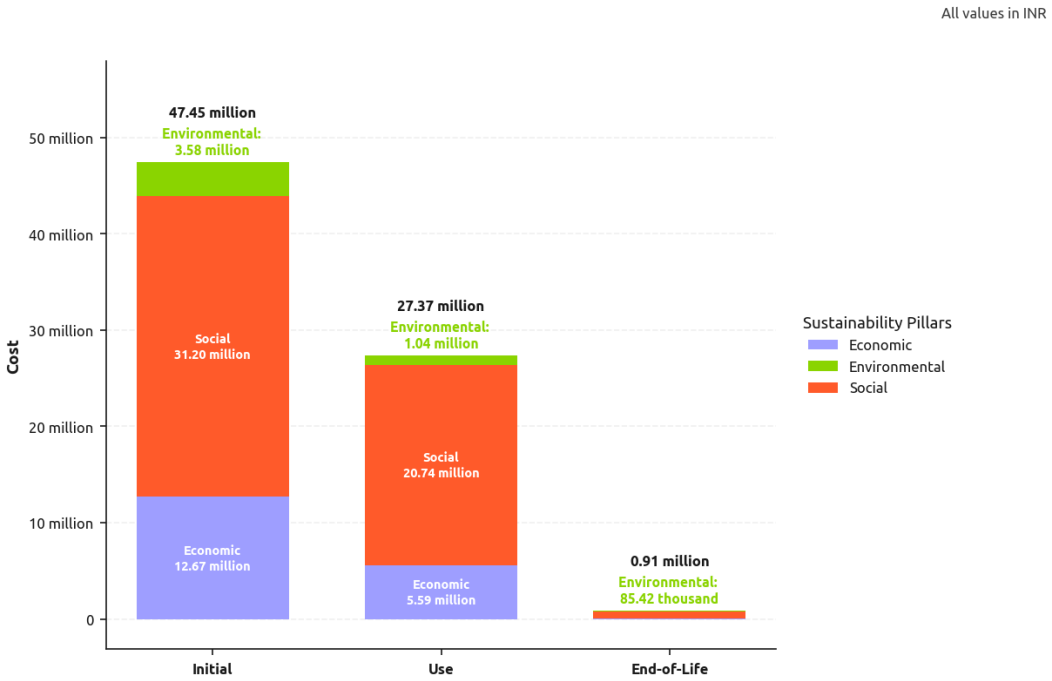


Figure 5. Distribution of various components of road user cost during construction

4 Summary and conclusions

The LCCA results indicate the relative contribution of construction, road user, and environmental costs, supporting informed and sustainable bridge planning decisions.

The most contributing stage of the life cycle is **Initial stage** contributing to around **62.66%** of the total life cycle cost.

The most contributing pillar is **Social** contributing to around **69.62%** of the total life cycle cost.

Appendix A: Assumptions

This Life Cycle Cost Assessment (LCCA) has been carried out using a deterministic, software-based approach. The analysis is based on standard engineering practice, published guidelines, and reasonable assumptions adopted to ensure consistency, transparency, and comparability of results. The key assumptions adopted in the present study are summarised below.

- The analysis period corresponds to the **design/service life of the bridge** as specified in the project data.
- The study includes: initial construction, routine inspection, periodic maintenance, major inspections, repairs, replacement of bearings and expansion joints, reconstruction, demolition and disposal, and associated road user costs.
- Only greenhouse gas (GHG) emissions are considered for environmental cost monetisation.
- Other impacts such as noise pollution, local business impacts, regional economic disruptions, and broader social externalities are outside the scope of this study.
- Components not explicitly defined in the input data or methodology are excluded.
- Future technological advancement, policy changes, traffic growth variations, or economic structural shifts are not considered.
- All costs are converted into present value terms.
- The discounting approach is based on standard economic theory where future costs and benefits are adjusted to present value using a constant discount rate.
- Inflation and interest rates are assumed constant over the analysis period.
- Real interest rate principles are used in deriving the discount relationship.
- Wholesale Price Index (WPI) ratios are used for escalation of specific cost components based on their respective commodity categories. WPI categories are used for adjusting the following components:
  - Passenger and crew cost → "All Commodities"
  - Property damage and spare parts → "Manufacture of parts and accessories for motor vehicles"
  - Human injury cost → "Medical accessories"
  - Petrol → "Mineral Oils"
  - Diesel → "HSD (High Speed Diesel)"
  - Engine oil → "Lube oils"
  - Grease and related oils → "Mineral oil (grouped)"
  - Tyre costs (by vehicle category) → Corresponding tyre index categories
  - Fixed and depreciation cost → "Manufacture of motor vehicles, trailers and semi-trailers"
  - Commodity holding cost → "Fuel & Power"
- Initial construction costs are based on estimated quantities and prevailing unit rates from SOR.
- Carbon emissions during construction are calculated using emission factors for material, transportation and energy consumption.
- Emission factors are assumed constant over the entire analysis period.
- Transportation of materials occurs under normal traffic and operating conditions.
- Maintenance and repair emissions are assumed proportional to initial construction emissions.
- Demolition and disposal emissions are assumed proportional to construction emissions.

- Recycling is assumed to generate cost recovery (treated as negative cost).
- Environmental cost is calculated by monetising carbon emissions using a single Social Cost of Carbon (SCC) value.
- Average Daily Traffic (ADT) is assumed constant within each defined analysis period.
- Traffic composition (cars, two-wheelers, buses, LCVs, HCVs, MCVs) is based on input data.
- Traffic diversion during construction, maintenance, repair, replacement, reconstruction, and demolition is assumed over a predefined detour length.
- Vehicle operating cost, value of time, and accident cost are calculated using standard equations and accepted parameters.
- Uniform traffic flow and average operating conditions are assumed during rerouting.
- Routine inspections, periodic maintenance, major inspections, and repairs occur at fixed predefined intervals.
- Maintenance and repair costs are calculated as a percentage of initial construction or superstructure cost.
- Replacement activities are limited to bearings and expansion joints.
- Reconstruction involves demolition of the existing bridge followed by construction of a new bridge with similar functional characteristics.
- All interventions occur as scheduled without unplanned delays unless explicitly stated.

Appendix B: Calculation methodology

This chapter presents the calculation method and equations used for cost calculations. The glossary for the terms used in the equations are mentioned below.

- A_{Di} = Accident distribution
- A_{Ne} = Number of accidents for each vehicle type
- A_{Tn} = Total number of accidents in the stipulated time
- C_{Di} = Distance related cost
- CF_D = Distance related congestion factor
- CF_T = Time related congestion factor
- C_{Ti} = Time related cost
- DC_m = Duration of construction in months
- D_{ma} = Number of equipment usage days
- D_{wm} = Number of working days in a month
- Di = Distance travelled for transportation of material
- EF_m = Emission factor for materials
- EF_{ma} = Emission factor for energy consumption
- EF_{tp} = Emission factor for transportation
- EF_v = Vehicular emission factor
- E_{VD} = Vehicle damage cost
- E_{Vi} = Economic cost of injury
- Q_{rm} = Quantity of recycle material
- RE_c = Recycling cost
- T_{AT} = Additional travel time
- T_{VP} = Time value
- V/C = Volume-Capacity ratio
- V_{CD} = Vehicle counts per day
- AC = Accident cost
- CC = Cargo capacity
- CFU = Price of fuel
- CHC = Commodity holding cost
- COF = Conversion factor
- CR = Crash rate
- CW = Crew cost
- DC = Depreciation cost
- D_{cf} = Demolition and disposal cost
- D_{cr} = Demolition and disposal cost for reconstruction
- DC_y = Duration of construction (years)
- DF_{EC} = Emission cost for end-of-life demolition
- DR_{EC} = Emission cost for demolition during reconstruction
- EOL = Engine oil consumption
- EOLC = Engine oil cost
- EOL_P = Engine oil price

- F_C = Fuel cost
- FC_{CB} = Fuel consumption petrol
- FC_{CS} = Fuel consumption diesel
- FL = Fall
- FXC = Fixed cost
- G = Grease consumption
- GC = Grease cost
- G_P = Grease price
- I = Interest rate
- IC = Initial construction cost
- IR = Investment ratio
- LC = Maintenance labour cost
- MIc = Major inspection cost
- MR_c = Major repair cost
- MR_{EC} = Major repair emission cost
- n = Design life (years)
- NP = New vehicle cost
- O_{Avg} = Average occupancy of a vehicle
- OL = Other oil consumption
- OLC = Other oil cost
- OL_P = Other oil price
- P_{IC} = Percentage of initial construction cost
- P_{IEC} = Percentage of initial carbon emission
- PMc = Periodic maintenance cost
- PM_{EC} = Periodic maintenance emission cost
- PT = Passenger time cost
- PWF = Present Worth Factor
- PWR = Power weight ratio
- Q = Quantity of material
- r = Discount rate
- R = Rate of material
- RC_{BE} = Replacement cost of bearing and expansion joint
- RCN = Reconstruction cost
- RD = Rerouting distance
- REC_{EC} = Reconstruction emission cost
- RG = Roughness
- RI_c = Routine inspection cost
- R = Rate of recycle material
- RS = Rise
- RUC = Road user cost
- SP = Spare part cost
- TC = Time cost
- TYC = Tyre cost
- TCe = Cost of each tyre
- TCR = Time cost for reconstruction

- TL = Tyre life
- TN = Number of tyres
- UPD = Utilisation per day
- VOC = Vehicle operating cost
- VOT = Value of time cost
- W = Width of carriageway
- WPI = Wholesale price index
- WZM = Work zone multiplier
- x = Periodic interval (years)
- ADT = Average daily traffic
- IEC = Initial emission cost
- IAEC = Initial carbon emission cost from on-site activities
- IETC = Initial carbon emission cost due to transportation of material
- SCC = Social cost of carbon
- VEC = Vehicular emission cost
- IAEC = Initial on-site emission cost
- ECR = Energy consumption rate
- H = Average machinery hours per day
- D_{ma} = Number of equipment usage days
- EF_{ma} = Emission factor for machinery energy use

B.1 Initial Cost

B.1.1 Economic cost

Initial construction cost

$$IC = \sum_{i=1}^{n} Q_i \times R_i$$

Time cost

$$TC = IC \times I \times DC_y \times IR$$

B.1.2 Social cost

Road user cost during construction

$$RUC_{cn} = VOC_{cn} + VOT_{cn} + AC_{cn}$$

$$VOC_{cn} = D_{wm} \times DC_m \times RD \times \left[\sum_{j=1}^{o} \left\{ (C_{Ti})_j \times (V_{CD})_j \times (CF_T)_j \times (WPI_{Ti})_j \right\} + \sum_{k=1}^{p} \left\{ (C_{Di})_k \times (V_{CD})_k \times (CF_D)_k \times (WPI_{Di})_k \right\} \right]$$

Distance related costs

$$F_{Cp} = CFU_{Pe} \times FC_{CS}$$

$$F_{Cd} = CFU_{Di} \times FC_{CB}$$

$$F_C = CFU \times FC$$

(Note: This table of equation is only for two lane road)

Table B-1. Fuel Consumption equations for different vehicles

Small and Big Car	$FC_{CS} = 30 + \frac{844.085}{V} + 0.003V^2 + (0.001 \times RG) + (0.3414 \times RS) - (0.2225 \times FL)$ $FC_{CB} = 35 + \frac{983.503}{V} + 0.003V^2 + (0.002 \times RG) + (0.339 \times RS) - (0.4785 \times FL)$
Two-wheeler	$FC = 2.704 + \frac{439.656}{V} + 0.00349V^2 + (0.000157 \times RG) + (0.3642 \times RS) - (0.2709 \times FL)$
Buses	$FC = 34.23 + \frac{4054.42}{V} + 0.02149V^2 + (0.001246 \times RG) + (3.4557 \times RS) - (1.8454 \times FL)$
LCV	$FC_{CB} = 22.504 + \frac{1708.244}{V} + 0.02591V^2 + (0.001612 \times RG) + (5.6863 \times RS) - (0.8744 \times FL)$
HCV	$FC_{CB} = 50.0 + \frac{8049.955}{V} + 0.012V^2 + (0.005 \times RG) + (4.565 \times RS) - (4.904 \times FL) - (7.285 \times PWR)$
MCV	$FC_{CB} = 90.0 + \frac{14489.919}{V} + 0.0216V^2 + (0.01 \times RG) + (8.217 \times RS) - (8.8272 \times FL) - (13.113 \times PWR)$

Table B-2. Speed equations for different vehicles

Small Car	$V = 81.19 - (0.7892 \times RF) - [0.001891 \times (RG - 2000)]$
Big Car	$V = 81.92 - (0.7963 \times RF) - [0.001915 \times (RG - 2000)]$
Two-wheeler	$V = 59.71 - (0.7892 \times RF) - [0.001891 \times (RG - 2000)]$
Buses	$V = 54.23 - (0.4111 \times RF) - [0.00098 \times (RG - 2000)]$
LCV	$V = 57.41 - (0.5119 \times RF) - [0.00102 \times (RG - 2000)]$
HCV	$V = 96.52 - (0.5040 \times RF) - [0.00100 \times (RG - 2000)]$
MCV	$V = 44.79 - (0.3994 \times RF) - [0.00079 \times (RG - 2000)]$

$$SP = \frac{SP}{NP} \times NP$$

Table B-3. Spare part to New vehicle cost ratio for different vehicles

Small Car	$\frac{SP}{NP} = 0.0075 \times (RG - 2000) \times 10^{-5}$
Big Car	$\frac{SP}{NP} = 0.0045 \times (RG - 2000) \times 10^{-5}$
Two-wheeler	$\frac{SP}{NP} = [-55.879 + (0.024 \times RG)] \times 10^{-5}$
Buses	$\frac{SP}{NP} = e^{-9.7871 + (0.007373 \times RF) + (0.0000723 \times RG) + \frac{1.925}{W}}$
LCV	$\frac{SP}{NP} = e^{-10.5615 + (0.000141 \times RG) + \frac{3.493}{W}}$
HCV and MCV	$\frac{SP}{NP} = e^{-9.492638 + (0.0001413 \times RG) + \frac{3.493}{W}}$

$$LC = a \times SP$$

a for small car and big car = 1.79934, Two-wheeler = 0.5498, Buses = 1.1781, LCV = 0.85773, HCV and MCV = 0.7912

$$TC = \frac{TC_e \times TN}{TL}$$

$$EOLC = EOL \times EOL_P \times 10^{-3}$$

$$OLC = OL \times OLP \times 10^{-4}$$

Table B-4. Tyre Life equations for different vehicles

Small and big car	$TL = 68771 - (147.9 \times RF) - (26.72 \times RG/W)$
Two-wheeler	$TL = 47340 - (101.8 \times RF) - (18.39 \times RG/W)$
Buses	$TL = 38519 - (389.52 \times RF) - (1.32 \times RG) + (983.829 \times W)$
LCV	$TL = 22382 - (375.3 \times RF) - (1.037 \times RG) + (3817 \times W)$
HCV	$TL = 24662 - (413.6 \times RF) - (1.142 \times RG) + (4205 \times W)$
MCV	$TL = 23726 - (398 \times RF) - (1.0099 \times RG) + (4046 \times W)$

Table B-5. Engine oil consumption for different vehicles

Small and big car	$EOL = 1.8807 + (0.036615 \times RF) + (0.000578 \times RG/W)$
Two-wheeler	$EOL = 0.405 + (0.007899 \times RF) + (0.000125 \times RG/W)$
Buses	$EOL = 0.4303 + (0.001494 \times RF) + (0.0007885 \times RG/W)$
LCV	$EOL = 0.80679 + (0.019496 \times RF) + (0.0001297 \times RG/W)$
HCV	$EOL = 1.0277 + (0.02495 \times RF) + (0.0001782 \times RG/W)$
MCV	$EOL = 1.3826 + (0.03348 \times RF) + (0.002319 \times RG/W)$

Table B-6. Other oil consumption equations for different vehicles

Small and big car and two-wheeler	$OL = 1.631 + (0.05167 \times RF) + (0.001867 \times RG/W)$
Buses	$OL = 3.3201 + (0.002889 \times RF) + (0.0008217 \times RG) - (0.3295 \times W)$
LCV	$OL = 2.0415 + (0.0001058 \times RG)$
HCV and MCV	$OL = 5.1037 + (0.0002646 \times RG)$

$$GC = G \times G_P \times 10^4$$

Table B-7. Grease consumption equations for different vehicles

Small car, big car and two-wheeler	$G = 2.816 + (0.2007 \times RF)$
Buses	$G = 4.992 + (0.03376 \times RF) + (0.3634 \times W)$
LCV	$G = 0.3661 + (0.0283 \times RF) + (0.000251 \times RG)$
HCV and MCV	$G = 0.9153 + (0.0707 \times RF) + (0.000627 \times RG)$

Time related costs

$$FXC = \frac{b}{UPD}$$

b for small car and big car = 395.65, Two-wheeler = 24.32, Buses = 772.89, LCV = 723.80, HCV = 924.28, MCV = 1238.26

$$DC = \frac{c}{UPD}$$

c for small and big cars = 42.83, Two-wheeler = 4.26, Buses = 221, LCV = 120.9, HCV = 154.54 and MCV = 238.54

Table B-8. Utilisation per day equations for different vehicles

Small car	$UPD = 6.7127 \times V$
Big Car	$UPD = 6.7378 \times V$
Two-wheeler	$UPD = 2.119 \times V$
Buses	$UPD = 22.7134 + (12.2569 \times V)$
LCV	$UPD = 28.807 + (2.1836 \times V)$
HCV	$UPD = 55.6719 + (4.22 \times V)$
MCV	$UPD = 77.7233 + (5.8915 \times V)$

For Car and Two wheelers,

$$PT = \frac{d}{V}$$

For Buses,

$$PT = \frac{d}{UPD}$$

d for small and big car = 328.06, Two-wheeler = 70.29, Buses = 15509.8

(Note: This equation is only for two lane road)

$$CW = \frac{e}{UPD}$$

e for Buses = 3775.3, LCV = 900, HCV = 1500 and MCV = 1800

$$CHC = \frac{f}{UPD}$$

f for LCV = 71.35, HCV = 218.75, MCV = 409.28

(Note: This equation is only for two lane road)

Time related congestion factors

Table B-9. Time related congestion factor equations for different vehicles

Cars	$CF_T = 1.087 + (0.483 \times V/C)$
Two-wheelers	$CF_T = 0.804 + (0.865 \times V/C)$
Buses	$CF_T = 0.864 + (0.543 \times V/C)$
LCV	$CF_T = 0.925 + (0.573 \times V/C)$
HCV and MCV	$CF_T = 0.878 + (0.561 \times V/C)$

(Note: This equation is only for two lane road)

Distance related congestion factor

(Note: This equation is only for two lane road)

$$VOT_{cn} = D_{wm} \times DC_m \times T_{AT} \times \left[\sum_{j=1}^{o} \{(T_{VP})_j \times (V_{CD})_j \times (O_{Avg})_j \times (WPI_{Ti})_j\} + \sum_{k=1}^{p} \{(CHC_T)_k \times (V_{CD})_k \times (WPI_{Di})_k\} \right]$$

Table B-10. Distance related congestion factor equations for different vehicles

Cars	$CF_D = 0.893 + (0.259 \times V/C)$
Two-wheelers	$CF_D = 0.917 + (0.112 \times V/C)$
Buses	$CF_D = 0.800 + (1.1 \times V/C)$
LCV	$CF_D = 0.9 + (1.0 \times V/C)$
HCV	$CF_D = 0.925 + (0.482 \times V/C)$
MCV	$CF_D = 0.900 + (1.4 \times V/C)$

$$AC_{cn} = A_{Tn} \times \left[\sum_{j=1}^{o} \left\{ (E_{Vi})_j \times (A_{Di})_j \times (WPI_{me})_j \right\} + \sum_{k=1}^{p} \left\{ (E_{VD})_k \times (A_{Ne})_k \times (WPI_{sp})_k \right\} \right]$$

$$A_{Tn} = D_{wm} \times DC_m \times CR \times WZM \times RD \times 10^{-6}$$

B.1.3 Environmental cost

- Embodied carbon emission cost during construction

$$IEC = SCC \times \sum_{j=1}^{o} [Q_j \times (COF)_j \times (EF_m)_j]$$

- Vehicular emission cost during construction due to rerouting

$$VEC = SCC \times D_{wm} \times DC_m \times RD \times \sum_{k=1}^{p} [ADT_k \times (EF_v)_k]$$

- Carbon emissions from on-site activities during construction

$$IAEC = SCC \times \sum_{j=1}^{o} [(ECR)_j \times (H)_j \times (D_{ma})_j \times (EF_{ma})_j]$$

- Carbon emission cost due to transportation of construction material

$$IETC = SCC \times \sum_{j=1}^{o} [Q_j \times Di_j \times (EF_{tp})_j]$$

B.2 Use Stage Cost

B.2.1 Economic cost

- **Routine Inspection cost**

$$RI_c = \text{PWF} \times P_{Ics}$$

$$\text{PWF} = \sum_{i=1}^{\text{int}\left(\frac{n}{x}\right)-1} \frac{(1+f)^{ix}}{(1+r)^{ix}}$$

- **Periodic maintenance cost**

$$PM_c = \text{PWF} \times P_{ICm}$$

- **Major repair cost**

$$MR_c = \text{PWF} \times P_{ICr}$$

- **Major inspection cost**

$$MI_c = \text{PWF} \times P_{ICi}$$

- **Replacement cost of bearings and expansion joints**

$$RC_{BE} = \text{PWF} \times P_{SCr}$$

B.2.2 Social cost

- **Road user cost due to rerouting during major repairs**

$$RUC_{mr} = VOC_{mr} + VOT_{mr} + AC_{mr}$$

$$VOC_{mr} = \text{PWF} \times D_{wm} \times DC_m \times RD \left[\sum_{j=1}^{o} \{(C_{Ti})_j \times (V_{CD})_j \times (CF_T)_j \times (WPI_{Ti})_j\} + \sum_{k=1}^{p} \{(C_{Di})_k \times (V_{CD})_k \times (CF_D)_k \times (WPI_{Di})_k\} \right]$$

(Refer to table B-1 to B-16 to calculate distance and time related costs for VOC)

$$VOT_{mr} = \text{PWF} \times D_{wm} \times DC_m \times T_{AT} \left[\sum_{j=1}^{o} \{(T_{VP})_j \times (V_{CD})_j \times (O_{Avg})_j \times (WPI_{Ti})_j\} + \sum_{k=1}^{p} \{(CHC_T)_k \times (V_{CD})_k \times (WPI_{Di})_k\} \right]$$

$$AC_{mr} = \text{PWF} \times A_{Tn} \left[\sum_{j=1}^{o} \{(E_{Vi})_j \times (A_{Di})_j \times (WPI_{me})_j\} + \sum_{k=1}^{p} \{(E_{VD})_k \times (A_{Ne})_k \times (WPI_{sp})_k\} \right]$$

- **Road user cost due to rerouting during major repairs**

$$RUC_{rbe} = VOC_{rbe} + VOT_{rbe} + AC_{rbe}$$

$$VOC_{rbe} = \text{PWF} \times D_{wm} \times DC_m \times RD \left[\sum_{j=1}^{o} \{(C_{Ti})_j \times (V_{CD})_j \times (CF_T)_j \times (WPI_{Ti})_j\} + \sum_{k=1}^{p} \{(C_{Di})_k \times (V_{CD})_k \times (CF_D)_k \times (WPI_{Di})_k\} \right]$$

(Refer to table B-1 to B-16 to calculate distance and time related costs for VOC)

$$VOT_{rbe} = \text{PWF} \times D_{wm} \times DC_m \times T_{AT} \left[\sum_{j=1}^{o} \{(T_{VP})_j \times (V_{CD})_j \times (O_{Avg})_j \times (WPI_{Ti})_j\} + \sum_{k=1}^{p} \{(CHC_T)_k \times (V_{CD})_k \times (WPI_{Di})_k\} \right]$$

$$AC_{rbe} = \text{PWF} \times A_{Tn} \left[\sum_{j=1}^{o} \{(E_{Vi})_j \times (A_{Di})_j \times (WPI_{me})_j\} + \sum_{k=1}^{p} \{(E_{VD})_k \times (A_{Ne})_k \times (WPI_{sp})_k\} \right]$$

B.2.3 Environmental cost

- Carbon emission cost due to periodic maintenance

$$PM_{EC} = \text{PWF} \times P_{IECp}$$

- Carbon emission cost due to major repairs

$$MR_{EC} = \text{PWF} \times P_{IECm}$$

- Carbon emission due to rerouting of vehicles during major repairs

$$VEC_{MR} = \text{PWF} \times \text{SCC} \times D_{wm} \times DC_m \times RD \times \sum_{k=1}^{p} [(ADT)_k \times (EF_v)_k]$$

- Carbon emission cost due to replacement of bearings and expansion joints

$$RC_{BEE} = \text{PWF} \times P_{IESr}$$

- Carbon emission due to rerouting of vehicles during replacement of bearings and expansion joints

$$VEC_{RE} = \text{PWF} \times \text{SCC} \times D_{wm} \times DC_m \times RD \times \sum_{k=1}^{p} [(ADT)_k \times (EF_v)_k]$$

B.3 End-of-Life Stage Cost Calculation

B.3.1 Economic cost

- **Demolition and disposal cost for reconstruction**

$$D_{cr} = \text{PWF} \times P_{ICd}$$

- **Reconstruction cost**

$$\text{RCN} = \text{PWF} \times \text{IC}$$

- **Time cost for reconstruction of bridge**

$$\text{TCR} = \left|\text{PWF} \times \text{TC}\right|$$

- **Demolition cost at end of life**

$$D_{cf} = \text{PWF} \times P_{ICd}$$

- **Recycling cost**

$$RE_c = \text{PWF} \times \sum_{i=0}^{m} (RS)_i \times (Q_{rm})_i$$

B.3.2 Social cost

- **Road user cost due to rerouting of vehicles during demolition for reconstruction**

$$\text{RUC}_{dr} = \text{VOC}_{dr} + \text{VOT}_{dr} + \text{AC}_{dr}$$

$$VOC_{dr} = \text{PWF} \times D_{wm} \times DC_m \times RD \left[\sum_{j=1}^{o} \{(C_{Ti})_j \times (V_{CD})_j \times (CF_T)_j \times (WPI_{Ti})_j\} + \sum_{k=1}^{p} \{(C_{Di})_k \times (V_{CD})_k \times (CF_D)_k \times (WPI_{Di})_k\}\right]$$

$$VOT_{dr} = \text{PWF} \times D_{wm} \times DC_m \times T_{AT} \left[\sum_{j=1}^{o} \{(T_{VP})_j \times (V_{CD})_j \times (O_{Avg})_j \times (WPI_{Ti})_j\} + \sum_{k=1}^{p} \{(CHC_T)_k \times (V_{CD})_k \times (WPI_{Di})_k\}\right]$$

$$AC_{dr} = \text{PWF} \times A_{Tn} \left[\sum_{j=1}^{o} \left\{ (E_{Vi})_j \times (A_{Di})_j \times (WPI_{me})_j \right\} + \sum_{k=1}^{p} \left\{ (E_{VD})_k \times (A_{Ne})_k \times (WPI_{sp})_k \right\} \right]$$

- **Road user cost due to rerouting of vehicles during reconstruction**

$$\text{RUC}_{re} = \text{VOC}_{re} + \text{VOT}_{re} + \text{AC}_{re}$$

$$VOC_{re} = \text{PWF} \times D_{wm} \times DC_m \times RD \left[\sum_{j=1}^{o} \left\{ (C_{Ti})_j \times (V_{CD})_j \times (CF_T)_j \times (WPI_{Ti})_j \right\} + \sum_{k=1}^{p} \left\{ (C_{Di})_k \times (V_{CD})_k \times (CF_D)_k \times (WPI_{Di})_k \right\} \right]$$

(Refer to table B-1 to B-16 to calculate distance and time related costs for VOC)

$$VOT_{re} = \text{PWF} \times D_{wm} \times DC_m \times T_{AT} \left[\sum_{j=1}^{o} \left\{ (T_{VP})_j \times (V_{CD})_j \times (O_{Avg})_j \times (WPI_{Ti})_j \right\} + \sum_{k=1}^{p} \left\{ (CHC_T)_k \times (V_{CD})_k \times (WPI_{Di})_k \right\} \right]$$

$$AC_{re} = \text{PWF} \times A_{Tn} \left[\sum_{j=1}^{o} \left\{ (E_{Vi})_j \times (A_{Di})_j \times (WPI_{me})_j \right\} + \sum_{k=1}^{p} \left\{ (E_{VD})_k \times (A_{Ne})_k \times (WPI_{sp})_k \right\} \right]$$

- **Road user cost due to rerouting of vehicles during demolition at the end of life**

$$\text{RUC}_{df} = \text{VOC}_{df} + \text{VOT}_{df} + \text{AC}_{df}$$

$$VOC_{df} = \text{PWF} \times D_{wm} \times DC_m \times RD \left[\sum_{j=1}^{o} \left\{ (C_{Ti})_j \times (V_{CD})_j \times (CF_T)_j \times (WPI_{Ti})_j \right\} + \sum_{k=1}^{p} \left\{ (C_{Di})_k \times (V_{CD})_k \times (CF_D)_k \times (WPI_{Di})_k \right\} \right]$$

(Refer to table B-1 to B-16 to calculate distance and time related costs for VOC)

$$VOT_{df} = \text{PWF} \times D_{wm} \times DC_m \times T_{AT} \left[\sum_{j=1}^{o} \left\{ (T_{VP})_j \times (V_{CD})_j \times (O_{Avg})_j \times (WPI_{Ti})_j \right\} + \sum_{k=1}^{p} \left\{ (CHC_T)_k \times (V_{CD})_k \times (WPI_{Di})_k \right\} \right]$$

$$AC_{df} = \text{PWF} \times A_{Tn} \left[\sum_{j=1}^{o} \left\{ (E_{Vi})_j \times (A_{Di})_j \times (WPI_{me})_j \right\} + \sum_{k=1}^{p} \left\{ (E_{VD})_k \times (A_{Ne})_k \times (WPI_{sp})_k \right\} \right]$$

B.3.3 Environmental cost

Carbon Emission Cost

- Carbon emission cost due to demolition and disposal for reconstruction

$$\text{DR}_{\text{EC}} = \text{PWF} \times P_{\text{IECd}}$$

- Carbon emission cost due to rerouting of vehicles during demolition for reconstruction

$$\begin{aligned} \text{VEC}_{\text{DR}} &= \text{PWF} \times \text{SCC} \times D_{wm} \times \text{DC}_m \times \text{RD} \\ &\times \textstyle\sum_{k=1}^{p} (\text{ADT})_k \, (\text{EF}_v)_k \end{aligned}$$

- Carbon emission cost due to reconstruction

$$\text{REC}_{\text{EC}} = \text{PWF} \times \text{IEC}$$

- Carbon emission cost due to rerouting of vehicles during reconstruction

$$\begin{aligned} \mathrm{VEC_{REC}} &= \mathrm{PWF} \times \mathrm{SCC} \times D_{wm} \times \mathrm{DC}_m \times \mathrm{RD} \\ &\times \sum_{k=1}^{p} (\mathrm{ADT})_k \, (\mathrm{EF}_v)_k \end{aligned}$$

- Carbon emission cost due to demolition and disposal at end of life

$$\mathrm{DF_{EC}} = \mathrm{PWF} \times P_{\mathrm{IECd}}$$

- Carbon emission cost due to rerouting of vehicles during demolition at end of life

$$\begin{aligned} \mathrm{VEC_{DF}} &= \mathrm{PWF} \times \mathrm{SCC} \times D_{wm} \times \mathrm{DC}_m \times \mathrm{RD} \\ &\times \sum_{k=1}^{p} (\mathrm{ADT})_k \, (\mathrm{EF}_v)_k \end{aligned}$$

Appendix C: Miscellaneous data

Table C-1. WPI Adjustment Factors – Combined. All values in INR.

Vehicle	Petrol Cost	Diesel Cost	Engine Oil Cost	Other Oil Cost	Grease Cost	Property Damage Cost	Tyre Cost	Spare Parts Cost	Fixed Depreciation	Commodity Holding Cost	Passenger Cost	Crew Cost	Fatal Injury Cost	Major Injury Cost	Minor Injury Cost	Value of Time Cost
Ratio (Selected / Base)																
Small Car	1.7210	1.7103	1.4497	1.6295	1.6295	1.1486	1.1266	1.1413	1.1464	1.4413	1.2783	1.2783	1.1055	1.1055	1.1055	1.2783
Big Car	1.7210	1.7103	1.4497	1.6295	1.6295	1.1486	1.1266	1.1413	1.1464	1.4413	1.2783	1.2783	1.1055	1.1055	1.1055	1.2783
Two Wheeler	1.7210	1.7103	1.4497	1.6295	1.6295	1.1486	1.1402	1.1413	1.1464	1.4413	1.2783	1.2783	1.1055	1.1055	1.1055	1.2783
Ordinary Bus	1.7210	1.7103	1.4497	1.6295	1.6295	1.1486	1.1926	1.2822	1.1464	1.4413	1.2783	1.2783	1.1055	1.1055	1.1055	1.2783
Deluxe Bus	1.7210	1.7103	1.4497	1.6295	1.6295	1.1486	1.1926	1.2822	1.1464	1.4413	1.2783	1.2783	1.1055	1.1055	1.1055	1.2783
LCV	1.7210	1.7103	1.4497	1.6295	1.6295	1.1486	1.1926	1.0099	1.1464	1.4413	1.2783	1.2783	1.1055	1.1055	1.1055	1.2783
HCV	1.7210	1.7103	1.4497	1.6295	1.6295	1.1486	1.1926	1.0099	1.1464	1.4413	1.2783	1.2783	1.1055	1.1055	1.1055	1.2783
MCV	1.7210	1.7103	1.4497	1.6295	1.6295	1.1486	1.1926	1.0099	1.1464	1.4413	1.2783	1.2783	1.1055	1.1055	1.1055	1.2783
Selected Year Values																
Small Car	146.97	161.45	190.20	150.73	150.73	130.02	111.76	129.19	130.46	146.58	154.93	154.93	146.48	146.48	146.48	154.93
Big Car	146.97	161.45	190.20	150.73	150.73	130.02	111.76	129.19	130.46	146.58	154.93	154.93	146.48	146.48	146.48	154.93
Two Wheeler	146.97	161.45	190.20	150.73	150.73	130.02	118.58	129.19	130.46	146.58	154.93	154.93	146.48	146.48	146.48	154.93
Ordinary Bus	146.97	161.45	190.20	150.73	150.73	130.02	116.28	145.14	130.46	146.58	154.93	154.93	146.48	146.48	146.48	154.93
Deluxe Bus	146.97	161.45	190.20	150.73	150.73	130.02	116.28	145.14	130.46	146.58	154.93	154.93	146.48	146.48	146.48	154.93
LCV	146.97	161.45	190.20	150.73	150.73	130.02	116.28	115.23	130.46	146.58	154.93	154.93	146.48	146.48	146.48	154.93
HCV	146.97	161.45	190.20	150.73	150.73	130.02	116.28	115.23	130.46	146.58	154.93	154.93	146.48	146.48	146.48	154.93
MCV	146.97	161.45	190.20	150.73	150.73	130.02	116.28	115.23	130.46	146.58	154.93	154.93	146.48	146.48	146.48	154.93
Base Year Values (2019)																
Small Car	85.40	94.40	131.20	92.50	92.50	113.20	99.20	113.20	113.80	101.70	121.20	121.20	132.50	132.50	132.50	121.20
Big Car	85.40	94.40	131.20	92.50	92.50	113.20	99.20	113.20	113.80	101.70	121.20	121.20	132.50	132.50	132.50	121.20
Two Wheeler	85.40	94.40	131.20	92.50	92.50	113.20	104.00	113.20	113.80	101.70	121.20	121.20	132.50	132.50	132.50	121.20
Ordinary Bus	85.40	94.40	131.20	92.50	92.50	113.20	97.50	113.20	113.80	101.70	121.20	121.20	132.50	132.50	132.50	121.20
Deluxe Bus	85.40	94.40	131.20	92.50	92.50	113.20	97.50	113.20	113.80	101.70	121.20	121.20	132.50	132.50	132.50	121.20
LCV	85.40	94.40	131.20	92.50	92.50	113.20	97.50	114.10	113.80	101.70	121.20	121.20	132.50	132.50	132.50	121.20
HCV	85.40	94.40	131.20	92.50	92.50	113.20	97.50	114.10	113.80	101.70	121.20	121.20	132.50	132.50	132.50	121.20
MCV	85.40	94.40	131.20	92.50	92.50	113.20	97.50	114.10	113.80	101.70	121.20	121.20	132.50	132.50	132.50	121.20

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